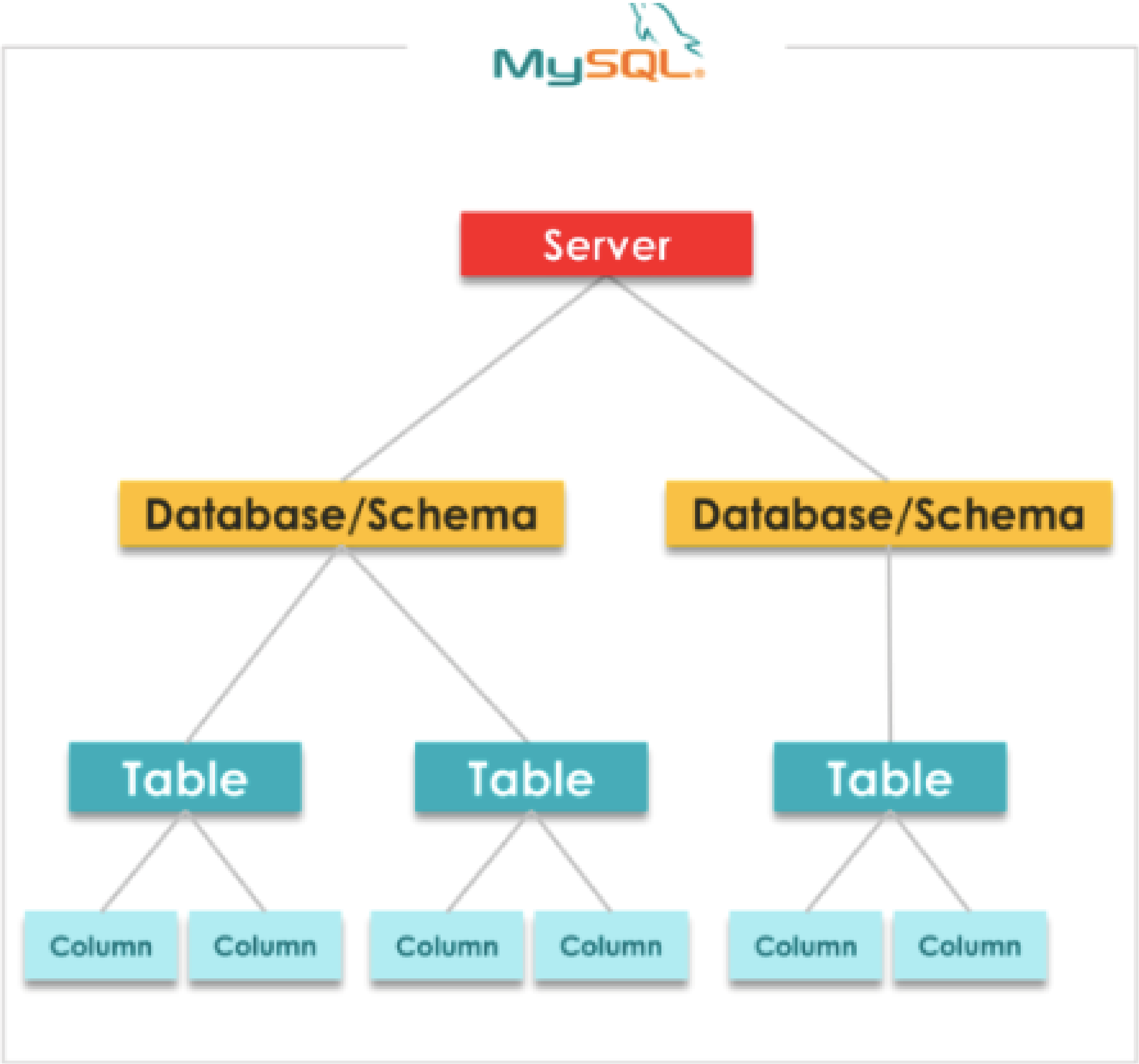
Introduction to SQL Commands

Objectives

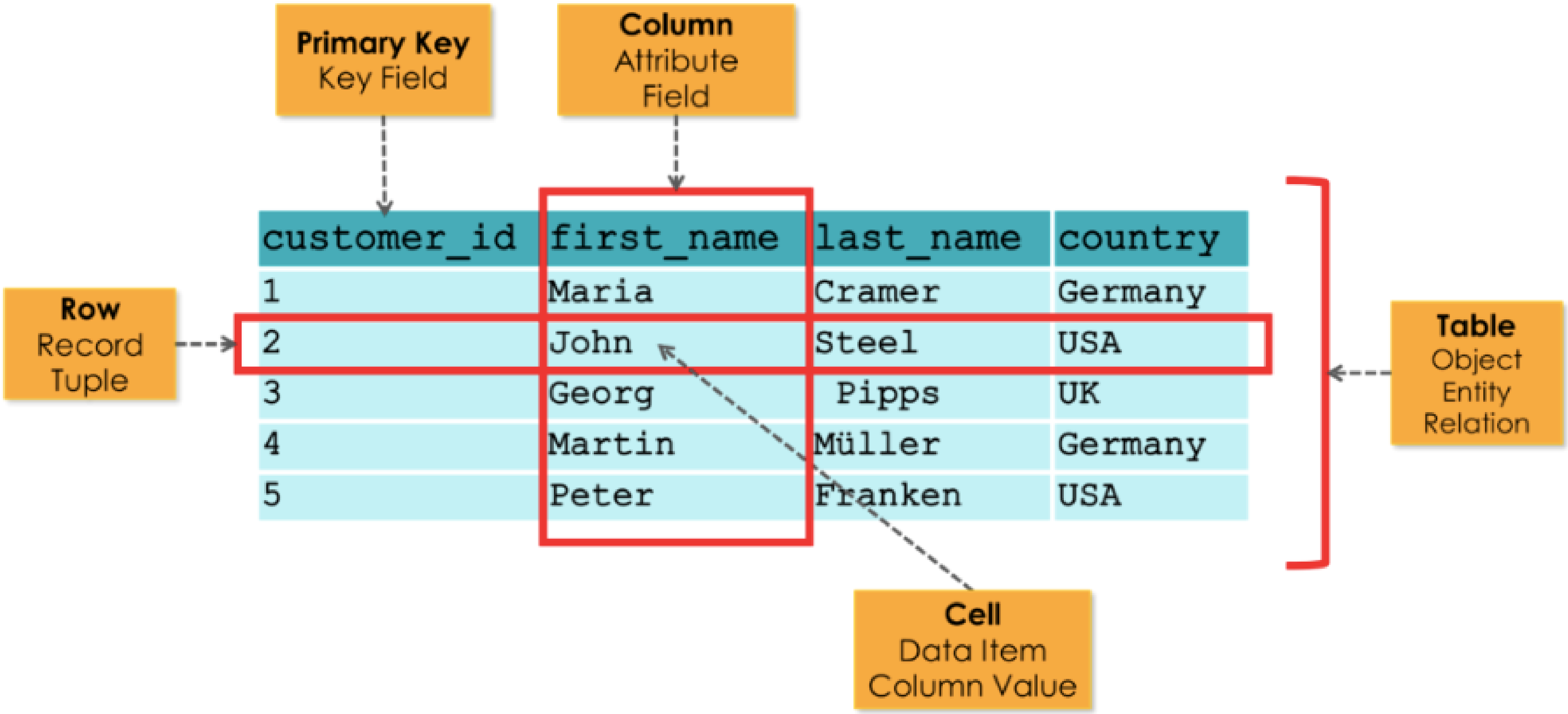
- Provide an overview of using MySQL Workbench
- Discuss SQL commands and its usage
- Know some of the suggested coding styles in SQL to make code readable
- Understand some of the fundamental SQL commands under DDL and DML.
- Demonstrate how to use some of the SQL commands under DDL and DML.

Important points to remember…

- In MySQL, schema and database are used interchangeably.



Important points to remember…

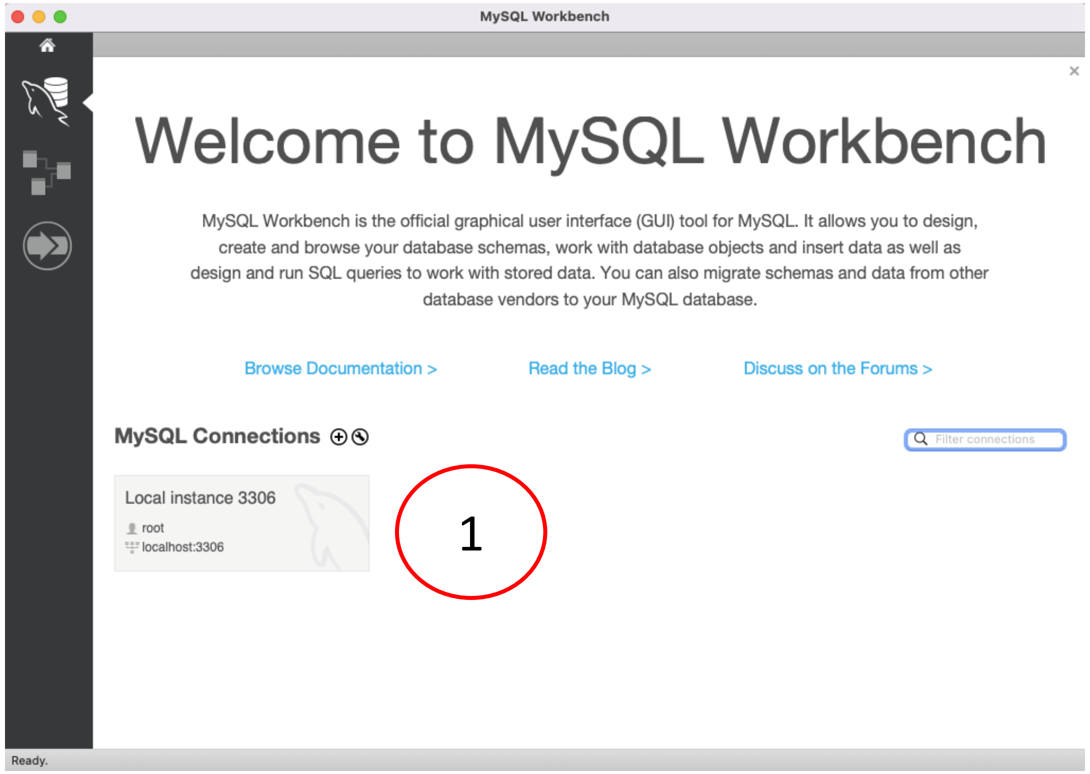


customer_id	first_name	last_name	country
1	Maria	Cramer	Germany
2	John	Steel	USA
3	Georg	Pipps	UK
4	Martin	Müller	Germany
5	Peter	Franken	USA

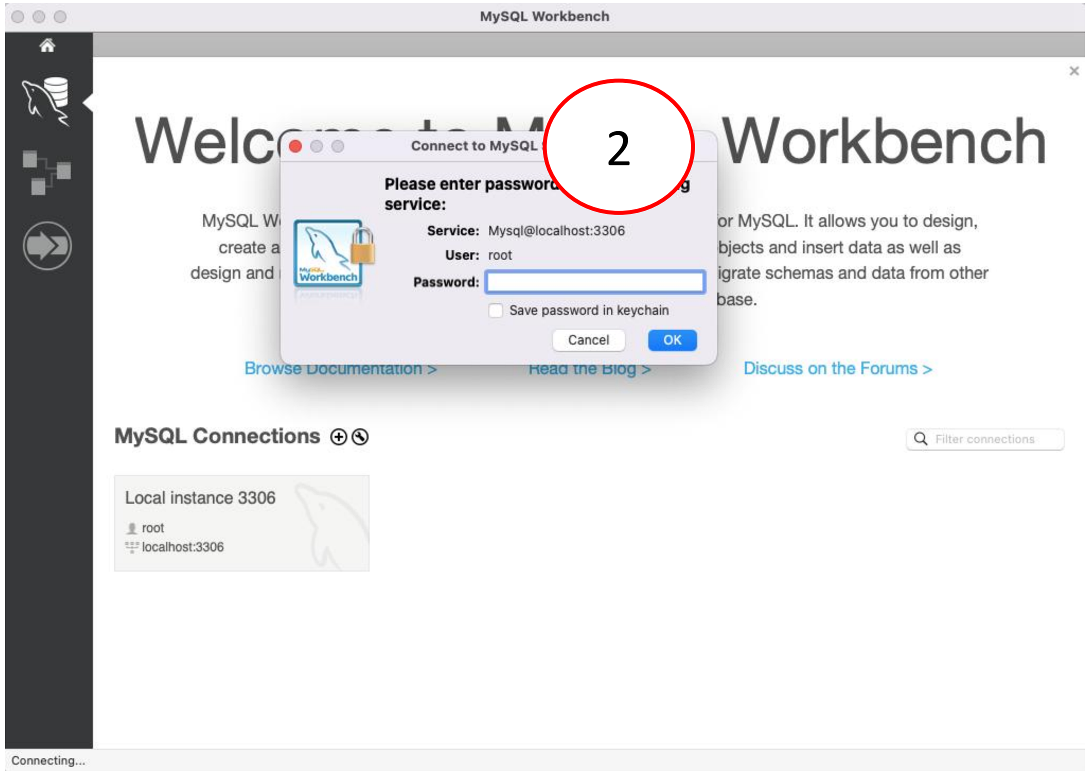
Important points to remember…

- Data Modeler: Entity and Attribute
- Database Developer: Table, Column, and Row
- Research Publications: Relation, Tuple, and Attribute

Overview of Using MySQL Server & Workbench



1 – Click on MySQL Conections



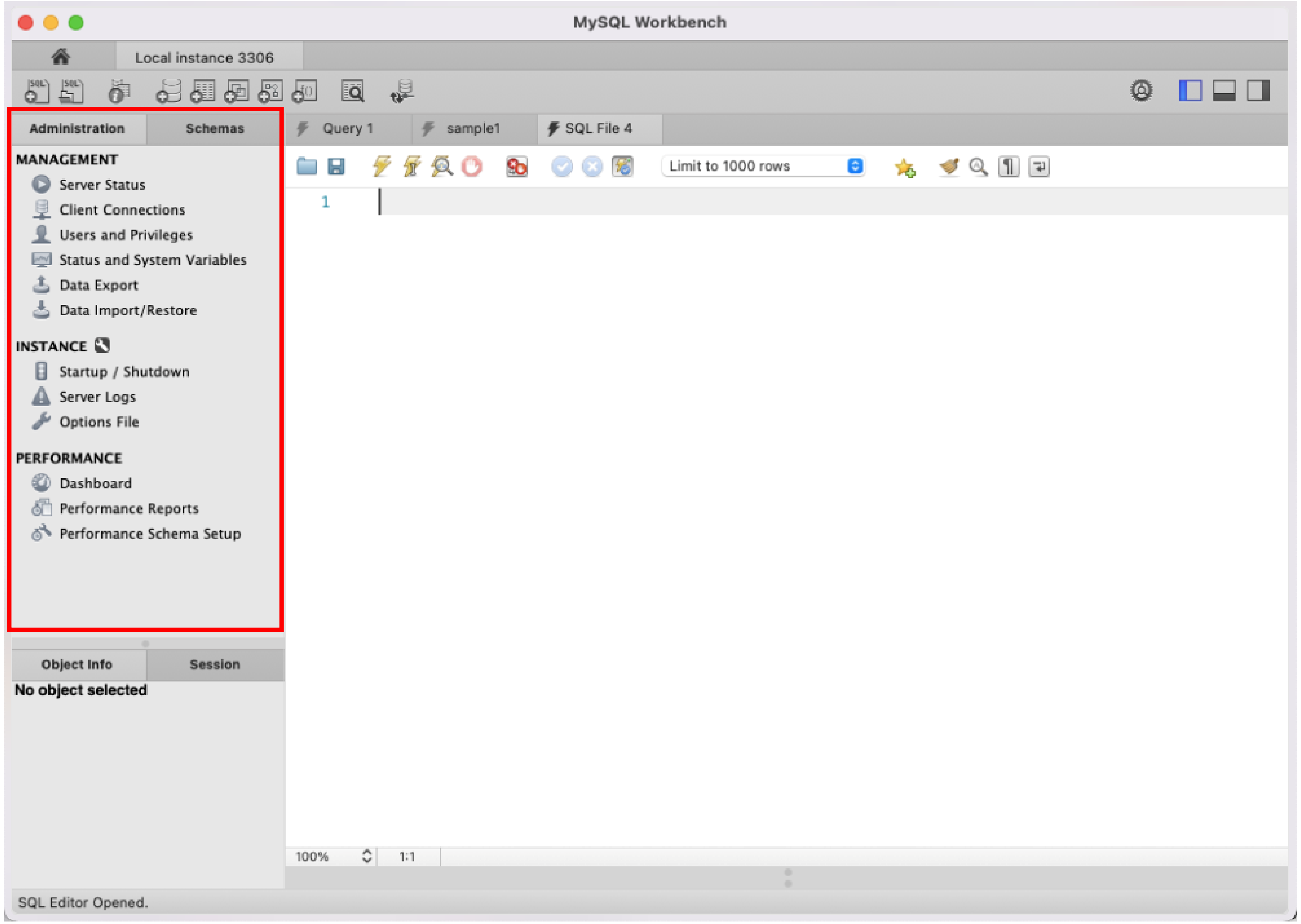
2- Input the password

Overview of Using MySQL Server & Workbench

1- Navigator contains two tabs Schemas and Administration:

•**Schemas** – allows you to navigate through your database objects.

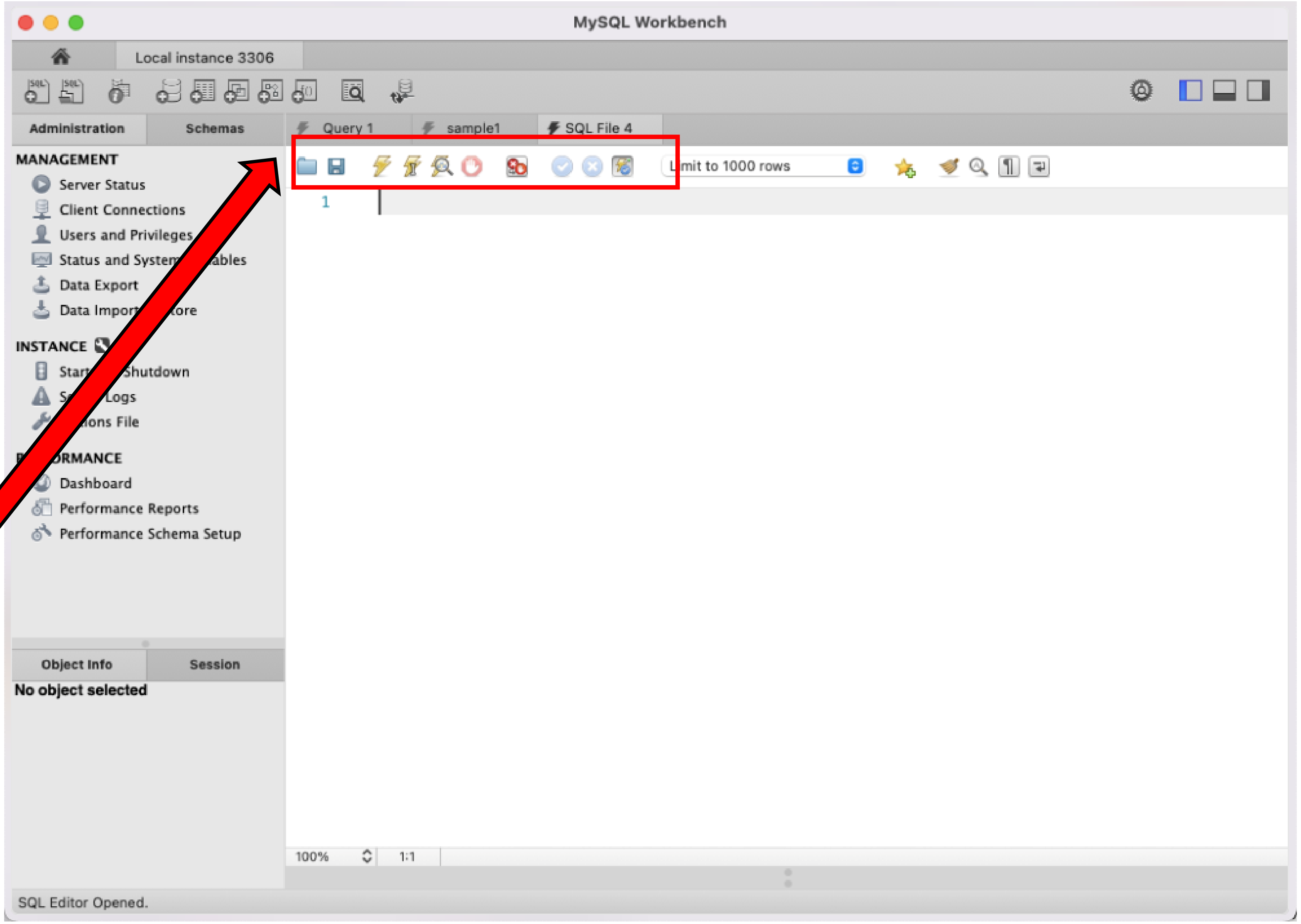
•**Administration** -contains all the tools for managing your SQL server, e.g. server status, users, starting and shutting down the server.



Overview of Using MySQL Server & Workbench

•**Main Toolbar** – contains the most frequently used functions in SQL, e.g. create a new SQL tab, create a new table, create a new schema. Contains menu tabs such as File, Edit, View, Query, etc.

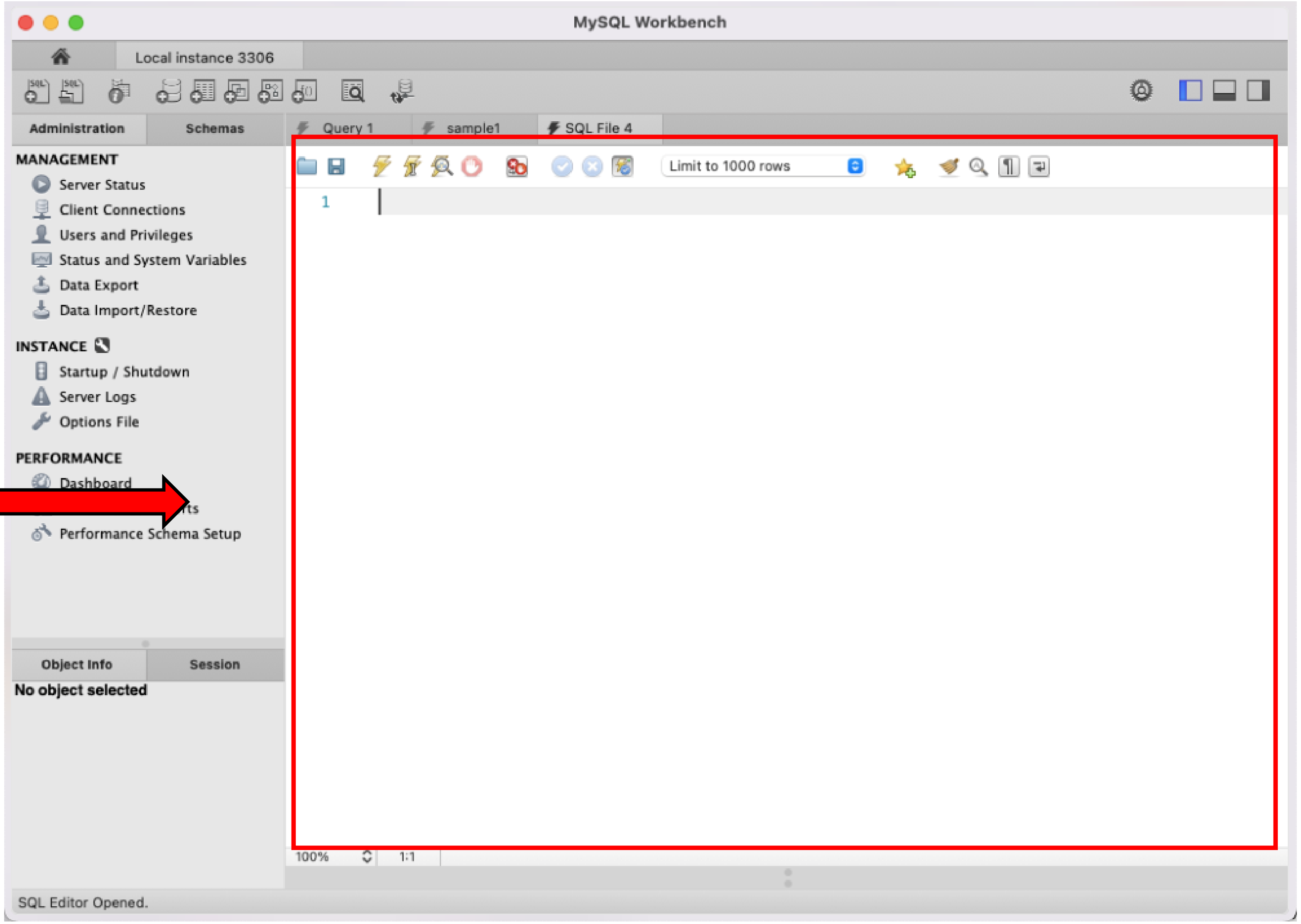
•**Query Toolbar** – provides actions for the SQL statement.



Overview of Using MySQL Server & Workbench

Query Editor

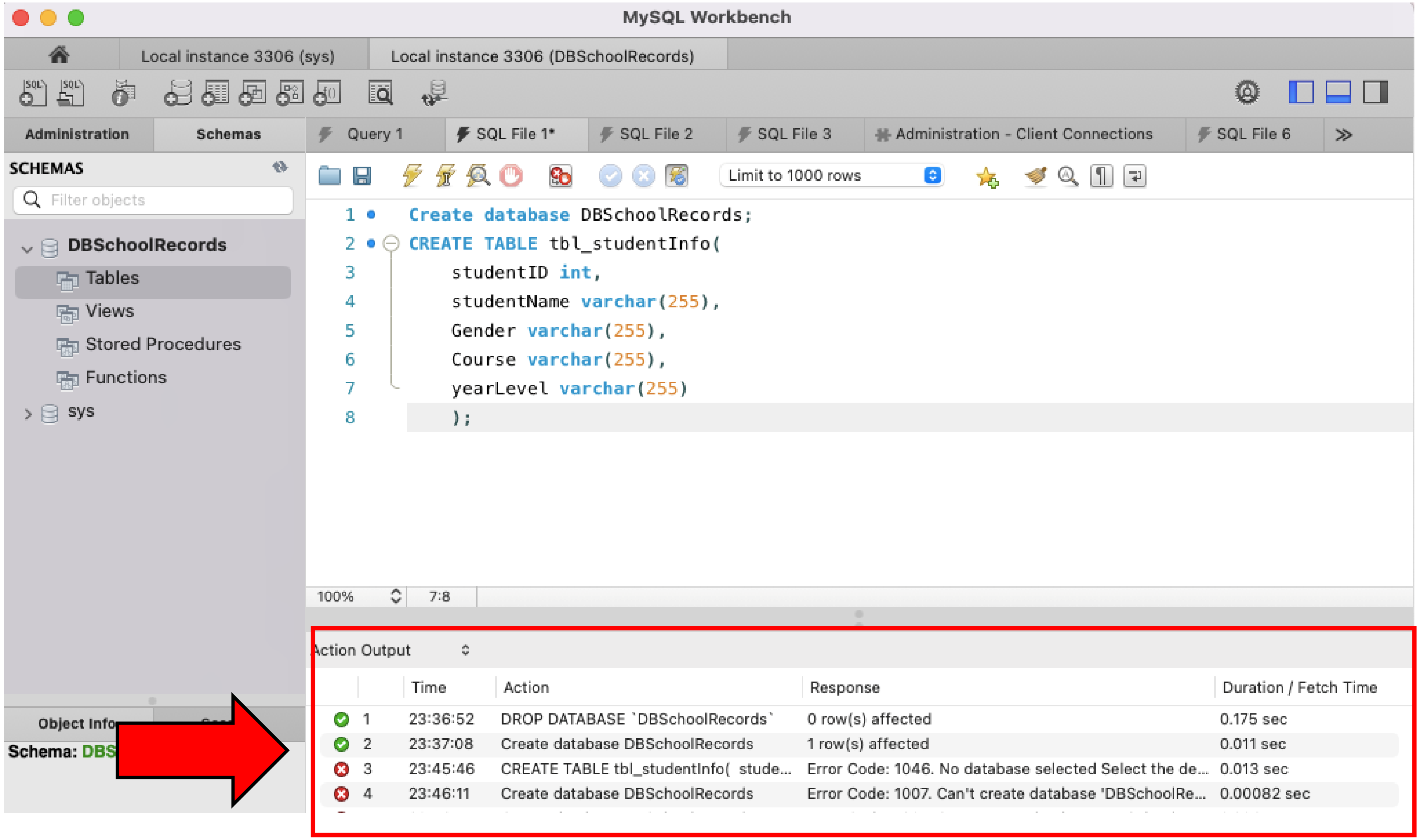
•Query Editor is the main place where you will write SQL statements. You can open as many tabs as you need.



Overview of Using MySQL Server & Workbench

Output Log

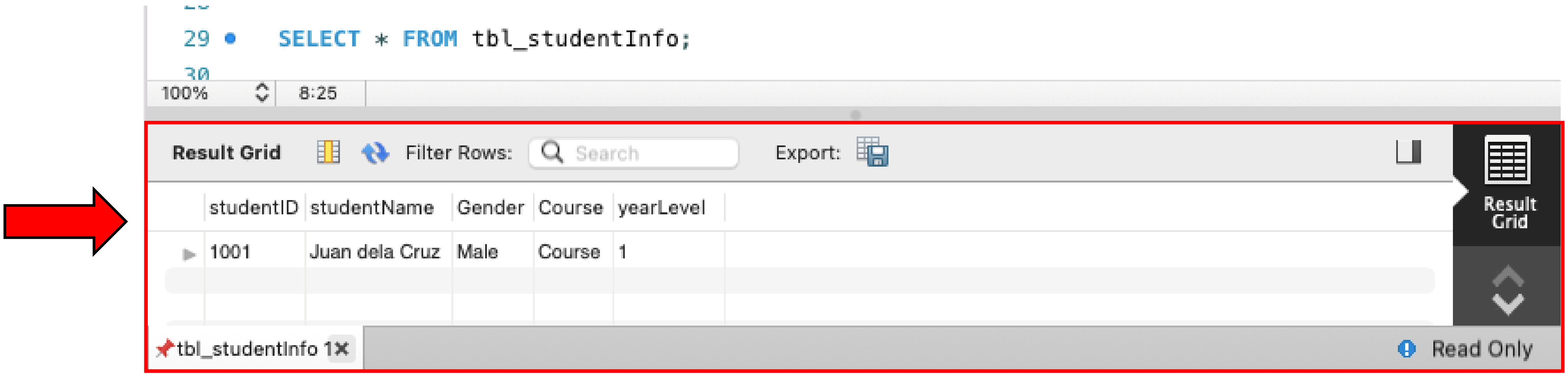
- Output (Log) is a section with information like action output, history, text output.



Overview of Using MySQL Server & Workbench

Result Grid

- Result Grid appears only after you have executed a **SELECT** statement. The data returned from the database is displayed in table form.



Overview of Using MySQL Server & Workbench

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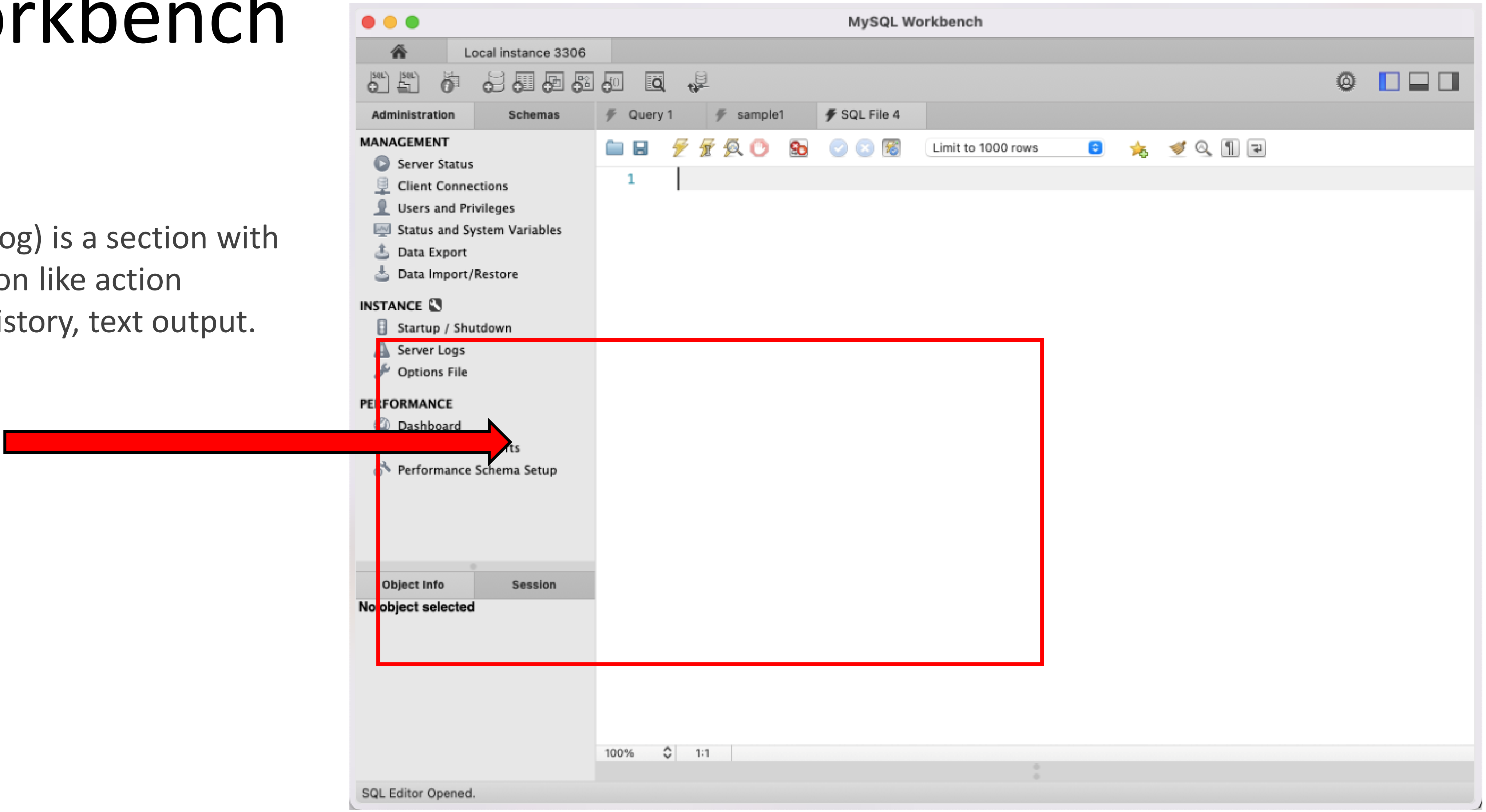


SQL Commands

- SQL commands are divided into different command groups depending on their purpose.
- You may find in other documentation that **SELECT** is considered a **DML** and not a **DQL**.

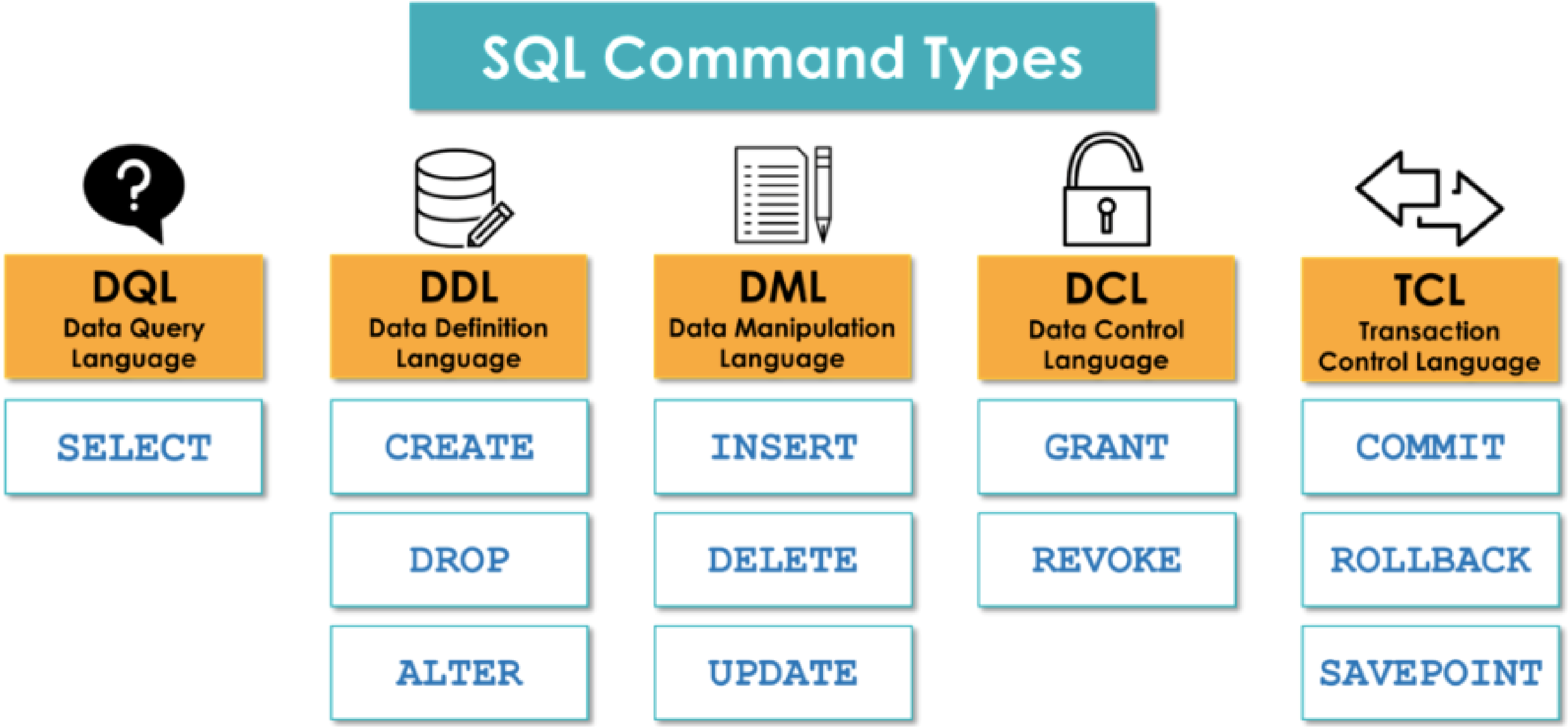


Photo reference: https://www.datawithbaraa.com/sql-introduction/sql-commands/

Data Definition Language (DML)

- **D**ata **D**efinition **L**anguage (**DDL**) is, as the name suggests, an SQL command that allows you to define, create and manipulate the structure of your databases such as creating or deleting tables, columns, views, or any other objects in your database.
- The following commands considered to be the main DDL commands:
- **CREATE** – creates new objects in the database.
- **DROP** – drops objects from the database.
- **ALTER** – alters the structure of objects in the database to be edited.

Data Query Language (DQL)

- **D**ata **Q**uery Language, (**DQL**), contains only one command called **SELECT**. It helps you to retrieve information from your database.
- **SELECT** is the most important command and concentrated focus of SQL. It contains many options and clauses which makes it very effective.

Data Manipulation Language (DML)

- **D**ata **M**anipulation **L**anguage, (**DML**), contains SQL commands used to manipulate your data inside your database.
- The **DML** command gives you tools to make changes to the records in a table, such as adding new records, deleting records, and updating records.
- The following commands considered to be the main DML commands:
- **INSERT** – inserts data into a table.
- **DELETE** – deletes data from a table.
- **UPDATE** – updates existing data in a table.

Data Control Language (DCL)

- **D**ata **C**ontrol **L**anguage, (**DCL**), contains SQL commands that allow you to control who can access your data in the database.
- Two general DCL commands are as following:
- **GRANT** – gives users access to the database.
- **REVOKE** – withdraws user access.

Transaction Control Language (TCL)

- **T**ransaction **C**ontrol **L**anguage (**TCL**), allows you to control and manage database transactions to maintain the integrity of your data.
- Here are the main TCL commands:
- **COMMIT** – save changes in the database.
- **ROLLBACK** – restore the database to the point of the last **COMMIT** or **SAVEPOINT** in case of any error.
- **SAVEPOINT** – create a point in a transaction to which you can later rollback.

DDL and DML Comparison

DDL	DML
Used to define database objects like tables, indexes, views, etc.	Used to manipulate data within the database.
Examples of DDL statements include CREATE, ALTER, and DROP.	Examples of DML statements include SELECT, INSERT, UPDATE, and DELETE.
Changes made using DDL affect the structure of the database.	Changes made using DML affect the data stored in the database.
DDL statements are not transactional, meaning they cannot be rolled back.	DML statements are transactional, meaning they can be rolled back if necessary.
DDL statements are usually executed by a database administrator.	DML statements are executed by application developers or end-users.
DDL statements are typically used during the design and setup phase of a database.	DML statements are used during normal operation of a database.

https://www.almabetter.com/bytes/tutorials/sql/dml-ddl-commands-in-sql

Suggested Coding Styles in SQL

Coding Style (SQL Coding Conventions)

- Clean code is code that is focused and understandable, which means it must be readable, logical, and changeable.
- Remember – good code is not the one computers understand; ***it is the one humans can understand.***

- *For sql coding best practices, visit this page:*

https://365datascience.com/tutorials/sql-tutorials/sql-best-practices/

Coding Style (SQL Coding Conventions)

- ***Which code is more readable?***

```sql
select c.customer_id,c.first_name,c.last_
from customers c join orders o on c.custo
where c.country='germany' AND o.quantity>
```

```sql
SELECT
    c.customer_id,
    c.first_name,
    c.last_name,
    o.order_id,
    o.order_date
FROM customers c
JOIN orders o
ON c.customer_id = o.customer_id
WHERE c.country = 'germany'
AND o.quantity > 100
```

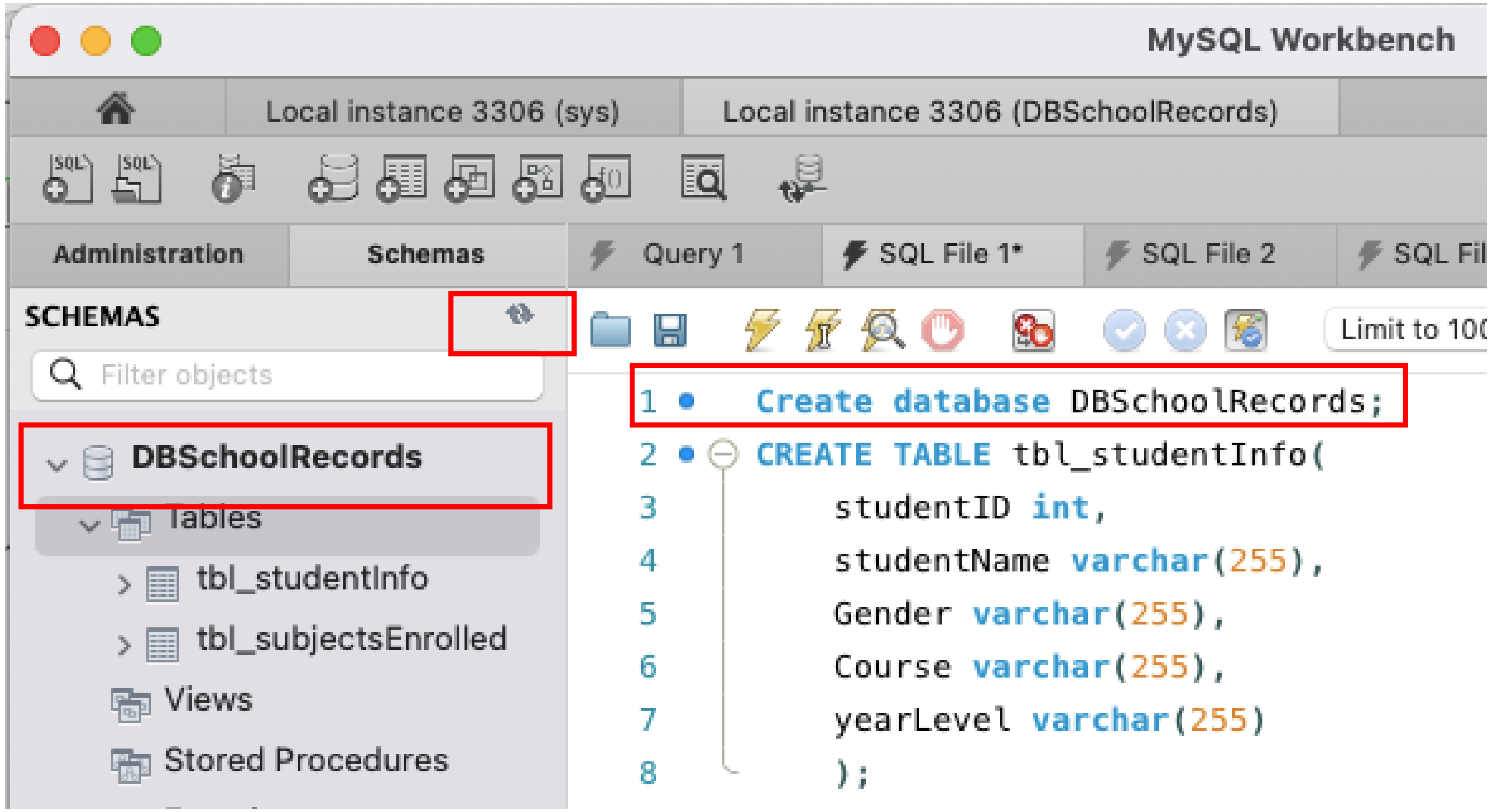
Photo reference: https://www.datawithbaraa.com/sql-introduction/sql-coding-style/

Suggested coding style

1. Add new lines for each keyword and selected column.
2. Make all keywords in caps to make them easier to find because each has its function and task.
3. You can add white space in WHERE and JOIN
4. Indentation makes your code more readable.

Creating a database in MySQL Workbench

- Using the CREATE DATABASE command, type the equivalent code. Click the refresh button in the navigator for the code to take effect.

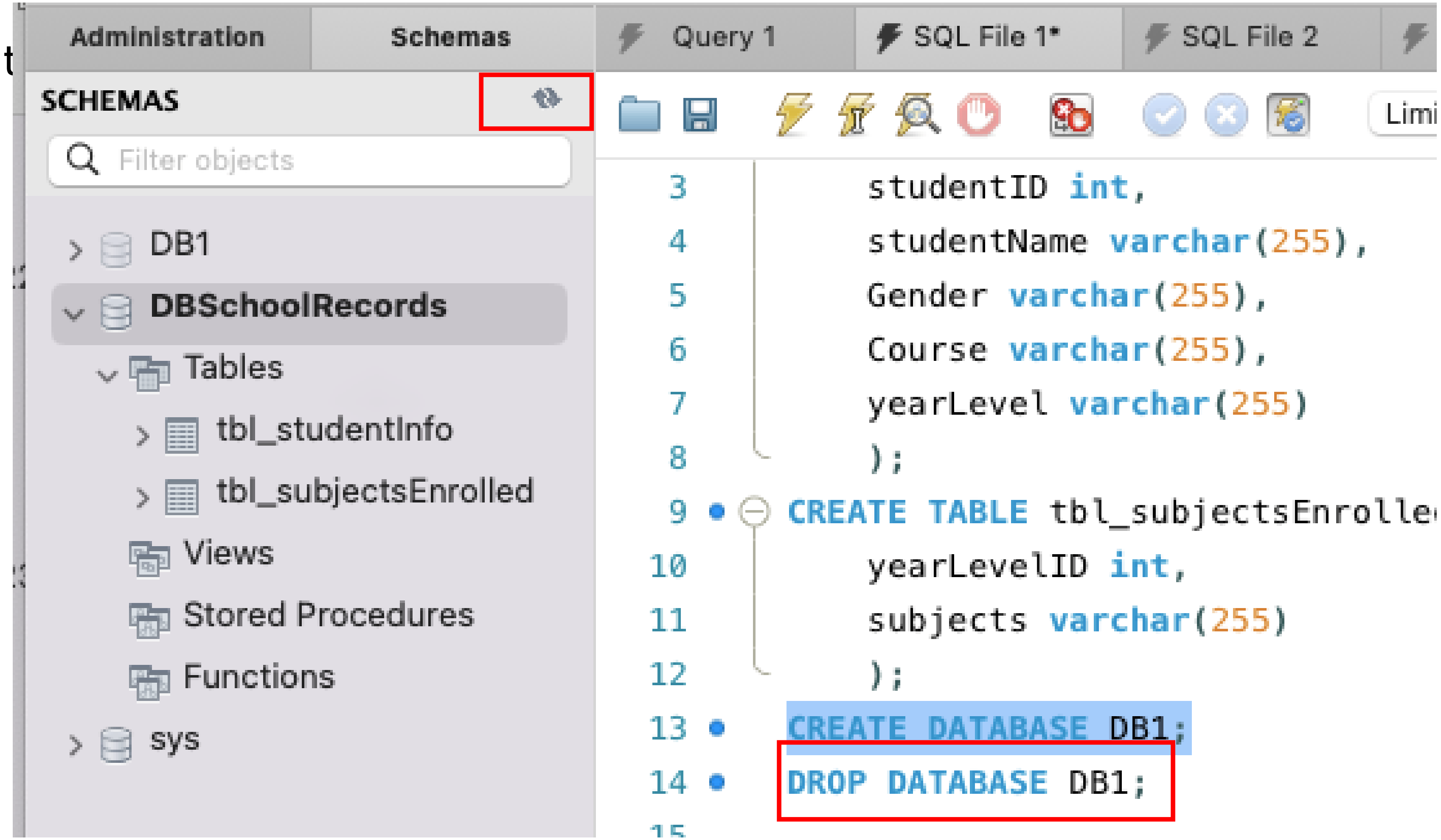


Dropping the database

- The DROP DATABASE statement is used to drop an existing SQL database.

Syntax:

```
DROP DATABASE databasename;
```



The MySQL CREATE TABLE Statement

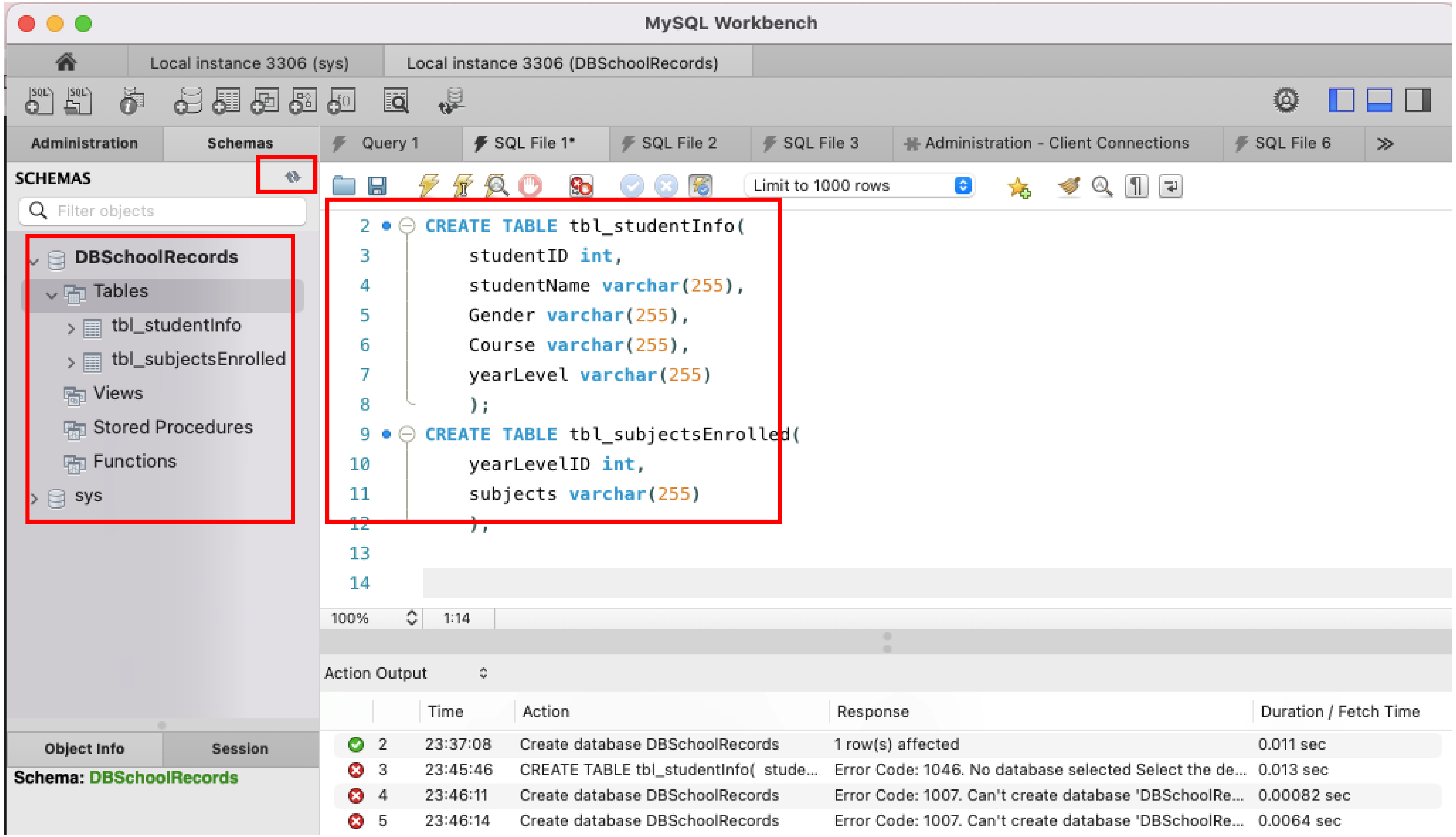
- The CREATE TABLE statement is used to create a new table in a database.

Syntax:

```
CREATE TABLE table_name (
    column1 datatype,
    column2 datatype,
    column3 datatype,
   ....
);
```

Creating tables in MySQL Workbench

Using the CREATE TABLE command, type the equivalent code. Click the refresh button in the navigator for the code to take effect.



The MySQL DROP TABLE Statement

- The DROP TABLE statement is used to drop an existing table in a database.

Syntax:

```
DROP TABLE table_name;
```

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Syntax:

```
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```

MySQL ALTER TABLE Statement

- The ALTER TABLE statement is used to add, delete, or modify columns in an existing table.
- The ALTER TABLE statement is also used to add and drop various constraints on an existing table.
-

ALTER TABLE - ADD Column

- To add a column in a table, use the following syntax:

```
ALTER TABLE table_name
ADD column_name datatype;
```

`Example:` The following SQL adds an "Email" column to the "Customers" table:

```
ALTER TABLE Customers
ADD Email varchar(255);
```

ALTER TABLE - DROP COLUMN

- To delete a column in a table, use the following syntax (notice that some database systems don't allow deleting a column):

```
ALTER TABLE table_name
DROP COLUMN column_name;
```

- `Example:`The following SQL deletes the "Email" column from the "Customers" table:

```
ALTER TABLE Customers
DROP COLUMN Email;
```

ALTER TABLE - MODIFY COLUMN

- To change the data type of a column in a table, use the following syntax:

```
ALTER TABLE table_name
MODIFY COLUMN column_name datatype;
```

```
EXAMPLE: (Adds a DateOfBirth column to the table
name Persons with data type: date)
```

```
ALTER TABLE Persons
ADD DateOfBirth date;
```

USING
SELECT STATEMENT

SQL Statement Query Command

SQL statement/Query/Command is used to perform tasks such as retrieve data or update data on databases.

https://www.datawithbaraa.com/sql-introduction/select-statement/

The MySQL SELECT Statement

- The SELECT statement is used to select data from a database.
- The data returned is stored in a result table, called the result-set.

The MySQL SELECT Statement

Syntax:

```
SELECT column1, column2, ...
FROM table_name;
```

Here, column1, column2, ... are the field names of the table you want to select data from. If you want to select all the fields available in the table, use the following syntax:

SELECT * FROM table_name;

The MySQL SELECT Statement

```
SELECT * FROM tbl_studentInfo;
```

studentID	studentName	Gender	Course	yearLevel
1001	Juan dela Cruz	Male	Course	1

Inserting Data

MySQL INSERT INTO Statement

- The INSERT INTO statement is used to insert new records in a table. It is possible to write the INSERT INTO statement in two ways:

1. Specify both the column names and the values to be inserted:

2. If you are adding values for all the columns of the table, you do not need to specify the column names in the SQL query. However, make sure the order of the values is in the same order as the columns in the table.

MySQL INSERT INTO Statement

Syntax:

- ```
  INSERT INTO table_name
  VALUES (value1, value2, value3, ...);
  ```
  ```

MySQL INSERT INTO Statement

- The following SQL statement inserts a new record in the "tbl_studentInfo" table:

```sql
INSERT INTO tbl_studentInfo (
studentID,
studentName,
Gender,
Course,
yearLevel)
VALUES (
'1001',
'Juan dela Cruz',
'Male',
'Course',
'1');
```

Activity:

- Open SQLyog
- Create a database. Database name: dbhr.
- Create a table inside the database: table name: tblEmployee
- Add the following data:

```
(001,'Alice Johnson','Software Engineer' ,'IT'               ,85,000 ,'Excellent'),
(002,'Bob Smith'    ,'Data Analyst'      ,'Data Science'     ,70,000 ,'Good'),
(003,'Carol White'   ,'HR Management'    ,'Human Resources'  ,95,000 ,'Excellent'),
(004,'David Brown'   ,'DevOps Engineer'  ,'IT'               ,88,000 ,'Satisfactiry'),
(005,'Eve Martin'    ,'Project Manager'  ,'Operations'       ,100,000,'Good'),
(006,'Frank Thomson' ,'Marketing Lead'   ,'Marketing'        ,78,000 ,'Excellent'),
(007,'Grace Williams','Sales Associate'  ,'Sales'            ,65,000 ,'Satisfactory'),
(008,'Henry Carter'  ,'UUUX Designer'    ,'IT'               ,75,000 ,'Excellent'),
(009,'Isla Johnson'  ,'Customer Support' ,'Support'          ,55,000 ,'Good'),
(010,'Jack Robinson' ,'Financial Analyst','Finance'          ,90,000 ,'Excellent');
```

Querying Data

References:

- https://www.w3schools.com/mysql/mysql_select.asp
- https://www.datawithbaraa.com/sql-introduction/sql-select/
- https://www.almabetter.com/bytes/tutorials/sql/dml-ddl-commands-in-sql